Physics Lab Report
Measurement of Gravitational Acceleration

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1 Introduction

The purpose of this experiment is to measure the gravitational acceleration using a simple pendulum. The theoretical formula for the period of a pendulum is:

$$T = 2\pi\sqrt{\frac{L}{g}} \tag{1}$$

where:

- T is the period,
- L is the length of the pendulum,
- g is the gravitational acceleration.

2 Method

The length of the pendulum was measured using a ruler. The time for 10 oscillations was recorded three times.

3 Results

3.1 Measured Data

Trial	Time for 10 Oscillations (s)	Period (s)
1	20.1	2.01
2	19.8	1.98
3	20.0	2.00

Table 1: Measured oscillation times

3.2 Calculation

From the formula:

$$g = \frac{4\pi^2 L}{T^2} \tag{2}$$

Substituting the measured values gives:

$$g \approx 9.8\,\mathrm{m/s^2} \tag{3}$$

4 Discussion

The measured value is close to the theoretical value of $9.81\,\mathrm{m/s^2}$. Small errors may have occurred due to reaction time.

5 Conclusion

The experiment successfully estimated the gravitational acceleration with reasonable accuracy.